Aceleração de uma partícula sobre uma superfície:

$$\frac{d^2q^i}{dt^2} + \Gamma^i{}_{jk}\frac{dq^j}{dt}\frac{dq^k}{dt} = 0$$

$$\Gamma^i{}_{jk} = g^{im}\left(g_{mj,k} + g_{mk,j} - g_{jk,m}\right)$$

Partícula Livre

$$\vec{a} = \frac{d\vec{v}}{dt} = 0 \qquad a = \frac{\partial v}{\partial q^i}\frac{dq^i}{dt} = \frac{\partial v}{\partial q^i} v^i$$

$$\vec{a} = \left(\frac{\partial v}{\partial q^i}\right)^k \vec{e}_k\, v^i \qquad a^k = V^k{}_{;i} V^i = 0$$

$$V^i V^k{}_{,i} + \Gamma^k{}_{mi} V^m V^i = 0$$

$$V^i \frac{\partial V^k}{\partial q^i} + \Gamma^k{}_{mi} V^m V^i = 0$$

$$V^i\left[\frac{\partial}{\partial q^i}\left(\frac{\partial q^k}{\partial t}\right)\right] = \frac{\partial q^i}{\partial t}\left[\frac{\partial}{\partial q^i}\left(\frac{\partial q^k}{\partial t}\right)\right] = \frac{\partial}{\partial t}\left(\frac{dq^k}{dt}\right)$$

$$\boxed{\ddot{q}^k + \Gamma^k{}_{mi} V^m V^i = 0}$$

Derivada Covariante da métrica:

$g_{ij} = \vec{e}_i \vec{e}_j$ e $q^i \to q^{i'}$ $\quad g_{i'j'} = \dfrac{\partial q^r}{\partial q^{i'}} \dfrac{\partial q^s}{\partial q^{j'}} g_{rs}$

Veremos que $g_{ij;k} = 0$ se $\Gamma^i{}_{jk} = \left\{ {i \atop jk} \right\} = \frac{1}{2} g^{im} (g_{mj,m} + g_{mk,j} - g_{jk,m})$

$$g_{ij;k} = g_{ij,k} - \Gamma^n{}_{ik} g_{nj} - \Gamma^m{}_{jk} g_{im} \qquad \frac{1}{2} g^{im} (g_{mj,m} + g_{mk,j} - g_{jk,m})$$

$$g_{ij;k} = g_{ij,k} - \frac{1}{2} g^{nm} (g_{mi,k} + g_{mk,i} - g_{ik,m}) g_{nj} - \frac{1}{2} g^{nm} (g_{mj,} + g_{mk,j} - g_{jk,m}) g_{in}$$

$$g_{ij;k} = g_{ij,k} - \frac{1}{2} \overbrace{g^{nm} g_{nj}}^{\delta^m{}_j} (g_{mi,k} + g_{mk,i} - g_{ik,m}) - \frac{1}{2} \underbrace{g^{nm} g_{in}}_{\delta^m{}_i} (g_{mj,k} + g_{mk,j} - g_{jk,m})$$

$$g_{ij;k} = g_{ij,k} - \frac{1}{2} (g_{ji,k} + \cancel{g_{jk,i}} - g_{ik,j}) - \frac{1}{2} (g_{ij,k} + g_{ik,j} - \cancel{g_{jk,i}})$$

$$g_{ij;k} = g_{ij,k} - \frac{1}{2}(2 g_{ij,k}) = 0$$

$$\boxed{g_{ij;k} = 0}$$

$$V_i = g_{ij} V^j \to V_{i;k} = (g_{ij} V^j)_{;k} = \cancel{g_{ij;k}}^{=0} V^j + g_{ij} V^j{}_{;k}$$

$$V_{i;k} = g_{ij} V^j{}_{;k}$$